Learning Objective 3.3: I can identify the hardware architecture of the ADALM-PHASER and control it via SDR software.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Walking the receive chain.** A wave from an X-band source arrives at the PHASER's patch face and ends up as numbers in the Raspberry Pi.

   (a) List, in order, the blocks the signal passes through from a single patch to the Raspberry Pi. Name the part number or device for each one.

   Patch element, then the ADL8107 LNA behind it, then the ADAR1000 analog beamformer, which applies phase and gain and sums its four elements. That sum goes to an LTC5548 mixer driven by the ADF4159 and HMC735 LO, which produces the 2.2 GHz IF. The IF goes to one receive channel of the ADALM-Pluto (AD9361), and the Raspberry Pi reads the IQ samples and serves the browser interface.

   (b) The ADAR1000 sets element phase with a 7-bit phase shifter. Compute the phase LSB in degrees, and state what physically limits how finely you can steer the beam.

   LSB = $\boxed{2.8125^\circ}$

   $$\text{LSB} = \frac{360^\circ}{2^7} = \frac{360^\circ}{128} = 2.8125^\circ$$

   The commanded phase is rounded to the nearest multiple of this step, so the applied phase ramp – and therefore the beam direction – can only take discrete values. The limit is the phase-shifter word length, not the software.

   (c) How many array elements feed one ADAR1000, and how many RF channels leave the pair of ADAR1000s on their way to the mixers?

   elements/chip = $\boxed{4}$

   RF channels = $\boxed{2}$

   Each ADAR1000 is a four-channel beamformer, so four of the eight elements feed each chip. Each chip sums its four inputs into one RF output, so two RF channels leave the pair – one per 4-element subarray.

   (d) The element spacing is $d = 14$ mm. For a source at 10.525 GHz, find the wavelength and the spacing in wavelengths.

   λ = $\boxed{28.5 \text{ mm}}$

   d/λ = $\boxed{0.491}$

   $$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{10.525 \times 10^9 \text{ Hz}} = 0.0285 \text{ m} = 28.5 \text{ mm}$$
   $$\frac{d}{\lambda} = \frac{14 \text{ mm}}{28.5 \text{ mm}} = 0.491$$

   Just under a half wavelength, which is the standard choice: close enough to $\lambda/2$ for a full scan range, but never above it, so no grating lobe enters visible space.

2. **The frequency plan.** The PHASER mixes the received X-band signal down to a fixed 2.2 GHz IF using high-side injection, so $f_{\mathrm{LO}} = f_{\mathrm{RF}} + f_{\mathrm{IF}}$. The ADF4159 and HMC735 VCO cover 12.2 to 13.0 GHz. The Pluto is tuned to 2.2 GHz and sampled at 3 MSPS.

   (a) **Find HB100** reports the lab source at 10.42 GHz. What LO frequency must the software command?

   $f_{\mathrm{LO}} =$ 12.62 GHz

   $$f_{\mathrm{LO}} = f_{\mathrm{RF}} + f_{\mathrm{IF}} = 10.42 + 2.20 = 12.62 \text{ GHz}$$

   This is inside the 12.2–13.0 GHz VCO range, so the source is reachable.

   (b) Given only the VCO limits and the fixed IF, what range of source frequencies can this receiver reach?

   $f_{\mathrm{RF,min}} =$ 10.0 GHz

   $$f_{\mathrm{RF}} = f_{\mathrm{LO}} - f_{\mathrm{IF}}: \quad \begin{aligned} &12.2 - 2.2 = 10.0 \text{ GHz} \\ &13.0 - 2.2 = 10.8 \text{ GHz} \end{aligned}$$

   $f_{\mathrm{RF,max}} =$ 10.8 GHz

   The receiver covers 10.0 to 10.8 GHz, which contains the whole 10.1 to 10.7 GHz spread of HB100 units.

   (c) A student skips **Find HB100**, so the software keeps the nominal LO of 12.725 GHz. The source in front of the array is actually at 10.30 GHz. What IF does the mixer produce, and by how much does it miss 2.2 GHz?

   $f_{\mathrm{IF}} =$ 2.425 GHz

   $\Delta f =$ 225 MHz

   $$\begin{aligned} f_{\mathrm{IF}} &= f_{\mathrm{LO}} - f_{\mathrm{RF}} = 12.725 - 10.300 = 2.425 \text{ GHz} \\ \Delta f &= 2.425 - 2.200 = 0.225 \text{ GHz} = 225 \text{ MHz} \end{aligned}$$

   The tone lands 225 MHz above where the Pluto is tuned.

   (d) Find the width of the baseband window the Pluto digitizes at 3 MSPS, and say in one or two sentences what the student of part (c) sees on the FFT tab.

   window = 3 MHz

   $$\text{window} = f_s = 3 \text{ MHz} \quad (\pm 1.5 \text{ MHz})$$

   The window is centred on the 2.2 GHz tuning. The offset of 225 MHz is 150 times the half-width of that window, so the tone is far outside it and nothing appears but the noise floor. An empty FFT with a powered, correctly aimed source is a frequency error, not a hardware failure.

3. **Hybrid beamforming.** The PHASER has eight elements, eight RF phase shifters, and two receive channels into the SDR.

   (a) Why does the board put a phase shifter behind every element at RF instead of digitizing all eight elements and doing the phase shift in software?

   An RF phase shifter and attenuator are small, cheap, and low-power, so eight of them cost little. A full receive channel – mixer, filter, ADC, and the data path to move the samples – is expensive in parts, board area, and power. Digitizing all eight elements would multiply the receiver hardware by four for a teaching board that mostly needs one beam at a time.

   (b) After the two 4:1 analog sums, how many independent complex weights can software still apply, and what information has been destroyed?

   digital weights = $\boxed{2}$

   Two, one per digital channel. Once four elements are summed in the ADAR1000 their individual amplitudes and phases no longer exist anywhere in the system; only the single sum survives. Software can weight and combine the two subarray outputs, but it cannot recover what any one element received.

   (c) An adaptive nulling algorithm running on the digital channels can steer roughly one null per available degree of freedom. State one capability the hybrid split keeps and one it takes away, and say which of the two you would trade if the board's job were to null several jammers at once.

   Kept: full per-element control of amplitude and phase at RF, so the array can steer, taper, and place a fixed null anywhere the designer computes in advance. Taken away: adaptive, data-driven nulling, because with two digital channels the adaptive processor has only one spare degree of freedom and can suppress about one interferer. To null several jammers at once you would trade the cheap analog architecture for more receive channels – ideally one per element – and accept the cost and power that come with it.

   (d) How many ADC channels would a fully digital 8-element version need, and name one practical cost besides part count.

   ADC channels = $\boxed{8}$

   Eight, one per element. Beyond the parts, the data rate off the board rises by a factor of four, the power budget grows with the added mixers and converters, and every channel needs its own amplitude and phase calibration before the samples can be combined coherently.

4. **Reading the bench.** Lab Preset 1 is loaded, the HB100 is at boresight about 1 m from the array, and the FFT tab is showing the baseband spectrum. The Pluto is tuned to 2.2 GHz and the LO is at 12.725 GHz.

   (a) The FFT peak sits 1 MHz above the centre of the window. Find the IF of the received tone and the frequency of the source.

   f_{IF} = $\boxed{2.201 \text{ GHz}}$

   $$f_{\mathrm{IF}} = 2.200 + 0.001 = 2.201 \text{ GHz}$$
   $$f_{\mathrm{RF}} = f_{\mathrm{LO}} - f_{\mathrm{IF}} = 12.725 - 2.201 = 10.524 \text{ GHz}$$

   f_{RF} = $\boxed{10.524 \text{ GHz}}$

   The source is 1 MHz below the nominal 10.525 GHz, which is ordinary for a free-running HB100.

   (b) **Rx Gain** is raised from 10 dB to 30 dB. Predict what happens to the peak level, to the noise floor, and to the separation between them, and explain why.

   Both the peak and the noise floor rise by close to 20 dB, and the separation between them changes very little. The Rx Gain is applied inside the AD9361, well downstream of the LNAs that already fixed how much noise accompanies the signal, so it amplifies signal and noise together. The one exception is the low end: at 10 dB the SDR's own noise and quantization floor make a measurable contribution, so the separation is slightly worse there than at 30 dB.

   (c) A student reports no peak anywhere in the FFT. Give the first two checks you would make, in order, and say what each one rules out.

   First, confirm the HB100 is powered and aimed at the array from about a metre – this rules out having no signal to receive at all, which is the most common cause. Second, run **Find HB100** and compare the result with the Signal Freq the GUI is using – this rules out a source sitting outside the 3 MHz baseband window because the LO is set for the wrong frequency. Only after both pass is it worth suspecting the array or the cabling.

   (d) On a uniform-taper beam sweep this board normally shows a noise floor about 23 dB below the peak. A student's sweep shows only 8 dB. Give two plausible causes and say how you would tell them apart.

   Either the received signal is too weak – source too far away, misaimed, or its battery low – so the peak has come down toward a fixed floor; or the array has not been calibrated, so the eight elements do not add coherently at broadside and the main lobe never reaches full height. To separate them, move the source closer and re-aim it: if the separation improves, it was signal level. If it does not, run **Calibrate** and sweep again.

Documentation: ______________________________